# A065-v0.3.8 | Source-Native Vorticity Core Closure

*Blind-compatible numerical preparation - NOT certified PKLSA physics*

## Scope and classification

These fields are GENERATED from original parametric trefoil centreline geometry using an ASSUMED Gaussian transverse smoothing width. They are 3D arrays, but they are not measurements or evolved Euler/PKLSA fields. Gamma and enstrophy inversion is conditional on the Gaussian cross-sectional hypothesis.

## Methods

Omega(x) = Gamma * integral over closed C of G_a(x-X(s)) dX(s), with G_a(R)=exp(-R^2/a^2)/(pi^(3/2) a^3). Compute divergence-free periodic FFT projection, velocity and pressure-Poisson diagnostics. For each local normal plane, compute Gamma = integral (omega dot tangent) dA and Z = integral (omega dot tangent)^2 dA. Gaussian hypothesis implies a = |Gamma|/sqrt(2 pi Z). A different model such as Rankine gives a different width.

## Numerical gates

| Gate | Status | Scope |
|---|---|---|
| G28 | PASS | 3D field / spectral constraints |
| G29 | FAIL | All transverse stations admissible? |
| G30 | PASS | At least one admissible inference per source |
| G31 | FAIL | No-fit withheld velocity and pressure predictions |
| G32 | UNRESOLVED | Authentic independently qualified PKLSA |

## Measured transverse profiles

**FOURIER_A** - a_seed = 0.500; periodic N=48^3; div(u) RMS=2.8e-18; Poisson residual=2e-13; projected source correction=0.023%.

| Station | Gamma | Z_axial | a(Gamma,Z) | Admitted | Vel error | Pressure error |
|---|---|---|---|---|---|---|
| 0.00 | 1.0661 | 0.5531 | 0.5719 | YES | 4.9% | 18.3% |
| 0.25 | 0.7278 | 0.7205 | 0.3421 | NO | 21.4% | 53.5% |
| 0.50 | -0.0026 | 1.2134 | 0.0009 | NO | 100.5% | 100.0% |
| 0.75 | 0.7278 | 0.7205 | 0.3421 | NO | 21.4% | 53.5% |

**TORUS_B** - a_seed = 0.650; periodic N=48^3; div(u) RMS=2.2e-18; Poisson residual=1.5e-18; projected source correction=0.034%.

| Station | Gamma | Z_axial | a(Gamma,Z) | Admitted | Vel error | Pressure error |
|---|---|---|---|---|---|---|
| 0.00 | 1.0992 | 0.3680 | 0.7228 | YES | 3.6% | 22.6% |
| 0.25 | 1.0701 | 0.3622 | 0.7094 | YES | 2.7% | 37.2% |
| 0.50 | 1.2963 | 0.4075 | 0.8101 | NO | 7.1% | 98.8% |
| 0.75 | 1.0701 | 0.3622 | 0.7094 | YES | 2.7% | 37.2% |

## Scientific interpretation / fail-closed limits

Some planes intersect neighbouring strands or contain strong non-axial vorticity, leading to cancellations. These are REJECTED before inference. G31 is a synthetic surrogate-prediction failure; the FFT-heldout velocity and

pressure derive from the same generated source and are NOT independently measured. The nonzero steady Euler residual shows the generated field is not established as an Euler steady state; no time evolution was solved.

For a physical claim the upstream PKLSA runner must export original u(x,y,z,t) or omega(x,y,z,t), a specified domain, boundary conditions, a transverse-core sampling region, circulation and enstrophy convergence, and a separate independently computed heldout u/p field with source hashes. The geometry-only E010/E011 outputs do not fulfil this requirement.